

Durgasantosh G

Software Engineer | Applied AI, Agentic Workflows, RAG, Python & Java

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PROFESSIONAL SUMMARY

AI-focused Software Engineer with 3+ years of experience building enterprise applications and applied AI products using **Python, Java, FastAPI, Spring Boot, React**, REST APIs, microservices, and cloud-native DevOps practices. Hands-on experience with **Google Agent Development Kit (ADK)**, LLM-based applications, agentic workflows, Retrieval-Augmented Generation (RAG), vector search, embeddings, semantic retrieval, LangChain, LangGraph, prompt engineering, context engineering, and AI evaluation. Built and deployed **rejekt.ai**, an AI-powered interview preparation platform with RAG, real-time voice workflows, and cloud deployment.

TECHNICAL SKILLS

Programming: Python, Java, TypeScript, JavaScript, SQL, C++, C#, HTML5, CSS3

AI Engineering: Google Agent Development Kit (ADK), LLM Application Development, AI Agents, Agentic Workflows, AI Agent Orchestration, Retrieval-Augmented Generation (RAG), LangChain, LangGraph, Model Context Protocol (MCP), Prompt Engineering, Context Engineering, Session Context Management, Embeddings, Semantic Search, Sentence-Transformers

LLM Platforms & ML: OpenAI, Google Gemini, NumPy, Pandas, Scikit-learn, spaCy, NLTK, TensorFlow, PyTorch, Statistical Analysis, Probability, Data Modeling, Data Visualization, Feature Engineering, AI Evaluation & Benchmarking

Backend & Data: FastAPI, Java 17, Spring Boot, Spring MVC, Spring Security, REST APIs, Microservices, JSON/XML, Authentication & Authorization, JWT, Kafka, RabbitMQ, PostgreSQL, MySQL, Oracle, SQL Server, MongoDB, Redis, DynamoDB, ChromaDB, pgvector (familiar)

Frontend & Cloud: React.js, Angular, Redux, GraphQL, AWS (EC2, S3, RDS, SQS, SNS, CloudWatch), Docker, Kubernetes, GitHub Actions, Jenkins, GitLab CI, CI/CD, Git, Maven, Postman, Agile/Scrum

PROFESSIONAL EXPERIENCE

Citi Group – New Jersey, USA

May 2025 – Present

Software Engineer

- Develop tool-enabled AI agent workflows using **Google Agent Development Kit (ADK)** and **Java 17**, enabling controlled orchestration of enterprise APIs, deterministic business rules, and multi-step workflow execution.
- Develop and enhance user-facing web applications using **React.js**, building reusable, responsive UI components for enterprise financial workflows.
- Design and integrate React components with **Java 17** and **Spring Boot** microservices, ensuring reliable frontend-backend communication and consistent data handling across distributed services.
- Develop and maintain secure **RESTful APIs** with validation, exception handling, **Spring Security**, and JWT-based authentication.
- Improve frontend performance by approximately **25%** through optimized API consumption, memoization, efficient state management, and reduced unnecessary re-renders.
- Support **Kafka**- and **DynamoDB**-based event-driven workflows, enabling reliable backend processing and low-latency updates for UI-driven applications.
- Participate in architecture and API contract discussions; diagnose production issues across frontend and backend services to improve reliability and release quality.
- Contribute to containerized deployments and **CI/CD** pipelines using **Docker**, **Kubernetes**, **Jenkins**, and **GitLab CI**.

Zensar Technologies – India

June 2021 – July 2023

Software Developer

- Delivered end-to-end features across **Angular 15**, **React.js**, **Java**, and **Spring Boot** for a microservices-based financial application.
- Developed secure **RESTful APIs** using Java and Spring Boot, implementing input validation, exception handling, and service-to-service integration.
- Built reusable UI components, client-side validations, and dynamic data-handling workflows, improving development efficiency by approximately **30%**.
- Integrated frontend modules with backend APIs, managing asynchronous requests, loading states, error handling, and responsive user interactions.
- Assisted with **Kafka**-based event-driven workflows for asynchronous transaction-related processing and contributed to PCI DSS-compliant financial modules.
- Participated in AWS deployments and CI/CD automation using **Jenkins** and **Docker** within Agile delivery cycles.

AI & MACHINE LEARNING PROJECTS

rejekt.ai – AI-Powered Interview Preparation Platform

www.rejekt.ai

- Designed and built an AI-powered interview preparation platform that generates personalized technical, behavioral, resume-based, job-specific, and company-specific mock interviews using **OpenAI** and **Google Gemini**.
- Developed end-to-end **RAG** workflows using **SentenceTransformers**, vector embeddings, **ChromaDB**, and semantic retrieval to produce context-aware interview questions and targeted feedback.
- Implemented a **LangGraph**-based adaptive interview workflow that evaluates responses for technical accuracy, STAR structure, filler words, communication quality, and missing concepts, then triggers targeted follow-up questions.
- Applied prompt and context engineering to manage multi-turn interview sessions, preserve relevant session context, and personalize recommendations based on prior responses.
- Integrated LLMs into full-stack workflows for interview generation, answer evaluation, candidate scoring, performance reporting, and conversational assistance, including tool/data integration patterns using **MCP**.
- Built real-time voice interview experiences using **Whisper Speech-to-Text**, Text-to-Speech, and automated audio-processing pipelines.
- Developed scalable backend services using **Python**, **FastAPI**, **PostgreSQL**, and **Redis**; deployed cloud-native services with **AWS S3**, **Docker**, **GitHub Actions**, and CI/CD pipelines.
- Built data pipelines to collect, normalize, deduplicate, and analyze job listings from ATS platforms and company career portals.

Beyond Sight: Empowering Visually Impaired Users with Audible Graphs – University of North Texas

ACM W4A '24 Research Paper

- Developed an AI-powered accessibility system that converts Excel charts and image-based graphs into structured audio narratives for visually impaired users.
- Implemented, evaluated, and benchmarked ResNet-based deep-learning models using **TensorFlow** and **PyTorch**, achieving **92% accuracy** in chart classification and data extraction.
- Engineered an OCR and NLP pipeline using **Pandas**, **NumPy**, **spaCy**, and **Tesseract** to extract chart titles, axes, legends, and values with **85% precision**.
- Integrated Text-to-Speech modules to provide real-time screen-reader-compatible audio feedback and improve chart interpretation speed by **60%**.
- Presented findings at **ACM W4A '24**, demonstrating applied ML, NLP, accessibility engineering, and research communication.

EDUCATION

University of North Texas – Dallas, Texas

August 2023 – May 2025

Master of Science in Computer Science

CERTIFICATIONS

AWS Certified Developer – Associate

Credential Verification